NATIONAL INSTITUTE OF TECHNOLGY PATNA
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
End Semester Examination July - Dec 2024

B. Tech: CSE I[st] semester Course Name: Programming in C
Course Code: CS14102 Max. Marks: 60 Time: 3 Hours

1. Write a program to find all unique pairs in an array that sum up to a given number k. **[5 M]|[CO4]|[BL3]**
Example : Input: Array: [1, 4, 6, 2, 3, 8], **k**: 7. **Output**: (1, 6), (4, 3)

2. Write a C program that swaps two numbers using pointers. The program should take two integers as input, and then swap their values using a function that accepts pointers as parameters. The function should NOT use any arithmetic operator (+,-,*,/) or third variable. **[5 M]|[CO7]|[BL3]**

3. Explain the concept of "pointer to pointer to pointer" in C. Write a C program that demonstrates the use of a pointer to a pointer to pointer to store and print the value of an integer using all the cases. **[1+4 = 5 M]|[CO7]|[BL2]**

4. Write a C program that demonstrates the usage of all types of operators in C. For each operator, explain its function and provide a simple example. **[5 M]|[CO2]|[BL1]**

5. Please go through the program below. List out all possible errors and warning. Fix those errors and warning to make the program work **properly**. Further, write output of the program assuming you provide your roll number to the program as input. **[6M]|[CO4,CO7]|[BL3]**

```
#include<stdio.h>
int main (){int rollNum;
printf("Enter your roll number"); scanf("%d",&rollNum);
char a[8]; a[8]= '\0';
for(int j = 0; j < 7; j++){ *a = rollNum % 10; a++; (*a)++; rollNum = rollNum / 10;}
for(int j = 0; j < 7; j++){printf("\n%d",a[j]);}
}
```

6. Guess the output of following programs assuming you input your roll number to the program. **[2+2 = 4 M|CO2|BL3]**

```
#include<stdio.h>
int main(){
unsigned int rollNum;
printf("Enter your roll number");
scanf("%u",&rollNum);
rollNum += (rollNum/100);
rollNum >>= (rollNum%3);
rollNum << (rollNum%3);
 printf("%u",rollNum);
}
```

```
#include<stdio.h>
int main(){ int rollNum;
printf("Enter your roll number");
scanf("%d",&rollNum);
rollNum = rollNum - 2*rollNum;
rollNum = rollNum%10;
printf("%d\n",rollNum);
if(rollNum){printf("1\n");}
rollNum=0;
if(rollNum){printf("\n2");}
rollNum=5;
if(rollNum){printf("3\n");}
}
```

7. Write a C program to take input your full name char by char and store it in a one dimensional array. Further, store content of your 1-D array in a 2 D array of size 10x10 **evenly**. If length of your name is not a perfect square, append 'X' char to make it a perfect square. For example, if the 2D array is of size 5x5 and your name is **Girish Kumar**, it should look as matrix below. Further make the program print the content of 2-D array in 2-D layout. Usage of any library function apart from printf and scanf is not allowed. **[10 M]|[CO4]|[BL3]**

G	I	R	I		
S	H		K		
U	M	A	R		
X	X	X	X		

OR

Describe the characteristics, scope, lifetime, and default values of the following storage classes. Write a C program to demonstrate the use of each storage class, showing how the value of a variable changes based on its storage class. Explain what a **void pointer** is and why it is used in C programming. Write a simple program to demonstrate the use of a void pointer to store and print the value of an integer, float and character variable. **[3+3+2+2=10 M]|[CO2,CO4,CO7]|[BL3]**

8. Will the following code snippets compile, if so what will be the output. Explain your answer in all cases. Assume that all necessary header files are already included. You should consider your Roll Number for the input to variable rollNum. **[2x10 = 20 M]**

a.[CO6][BL3]

```
void function() {
    int x = 5;
    x++;
    printf("%d ", x);
}
int main() {
    unsigned int rollNum;
    printf("Enter your roll number");
    scanf("%u",&rollNum);
    int x = 0;
    int y = rollNum%10? rollNum%10:3;
    for (int i = 0; i < y; i++) {
        function();
    }
    return 0;
}
```

b.[CO2][BL3]

```
int main()
{
    unsigned int rollNum;
    printf("Enter your roll number");
    scanf("%u",&rollNum);
    int y = rollNum%10? rollNum%10:3;
    int x = 5;
printf("x << 1: %d\n", x << 1);
printf("y >> 1: %d\n", y >> 1);
return 0;
}
```

c.[CO4][BL3]

```
int main(){
char a[] = {'a','b','c','\0','d','e','\0'};
printf("%s",a);
}
```

d.[CO3][BL3]

```
int main()
{
    unsigned int rollNum;
    printf("Enter your roll number");
    scanf("%u",&rollNum);
    int y = rollNum%10? rollNum%10:5;
    int sum = 0;
for (int i = 1; i <= y; i++)
{
  sum += i * i;
}
printf("Sum of squares: %d\n", sum);
return 0;
}
```

e.[CO2][BL3]

```
int main()
{
    unsigned int rollNum;
    printf("Enter your roll number");
    scanf("%u",&rollNum);
    int y = rollNum%10? rollNum%10:7;
int x = 3;
printf("x == y: %d\n", x == y);
printf("x < y && y > 5: %d\n", x < y && y > 5);
printf("x & y: %d\n", x & y);
return 0;
}
```

f.[CO2][BL3]

```
int main()
{
    unsigned int rollNum;
    printf("Enter your roll number");
    scanf("%u",&rollNum);
    int x = 0;
    int b = rollNum%10? rollNum%10:8;
int a =5;

int result = a++ + ++b;
printf("a = %d, b = %d, result = %d\n", a, b, result); return
0;
}
```

g.[CO2][BL3]

```
int main() {
    unsigned int rollNum;
    printf("Enter your roll number");
    scanf("%u",&rollNum);
    int a = rollNum%10? rollNum%10:5;
    int b = 10;
    printf("a == b: %d\n", a == b);
    printf("a != b: %d\n", a != b);
    printf("a < b: %d\n", a < b);
    printf("a > b: %d\n", a > b);
    return 0;
}
```

h.[CO2][BL3]

```
int main() {
    unsigned int rollNum;
    printf("Enter your roll number");
    scanf("%u",&rollNum);
    int a = rollNum%2? rollNum%2:1;
    int  b = 0;
    printf("a && b: %d\n", a && b);
    printf("a || b: %d\n", a || b);
    printf("!a: %d\n", !a);
    return 0;
}
```

i.[CO2][BL3]

```
int main() {
    unsigned int rollNum;
    printf("Enter your roll number");
    scanf("%u",&rollNum);
    int b = rollNum%10? rollNum%10:3;
    int a = 5;
    printf("a & b: %d\n", a & b);  // Bitwise AND
    printf("a | b: %d\n", a | b);  // Bitwise OR
    printf("a ^ b: %d\n", a ^ b);  // Bitwise XOR
    return 0;
}
```

j.[CO3][BL3]

```
int main() {
    int a;
    int result = scanf("%d", &a);
    if (result == 1) {
        printf("Input successful! You entered: %d\n", a);
    } else {
        printf("Input failed.\n");
    }
    return 0;
}
```